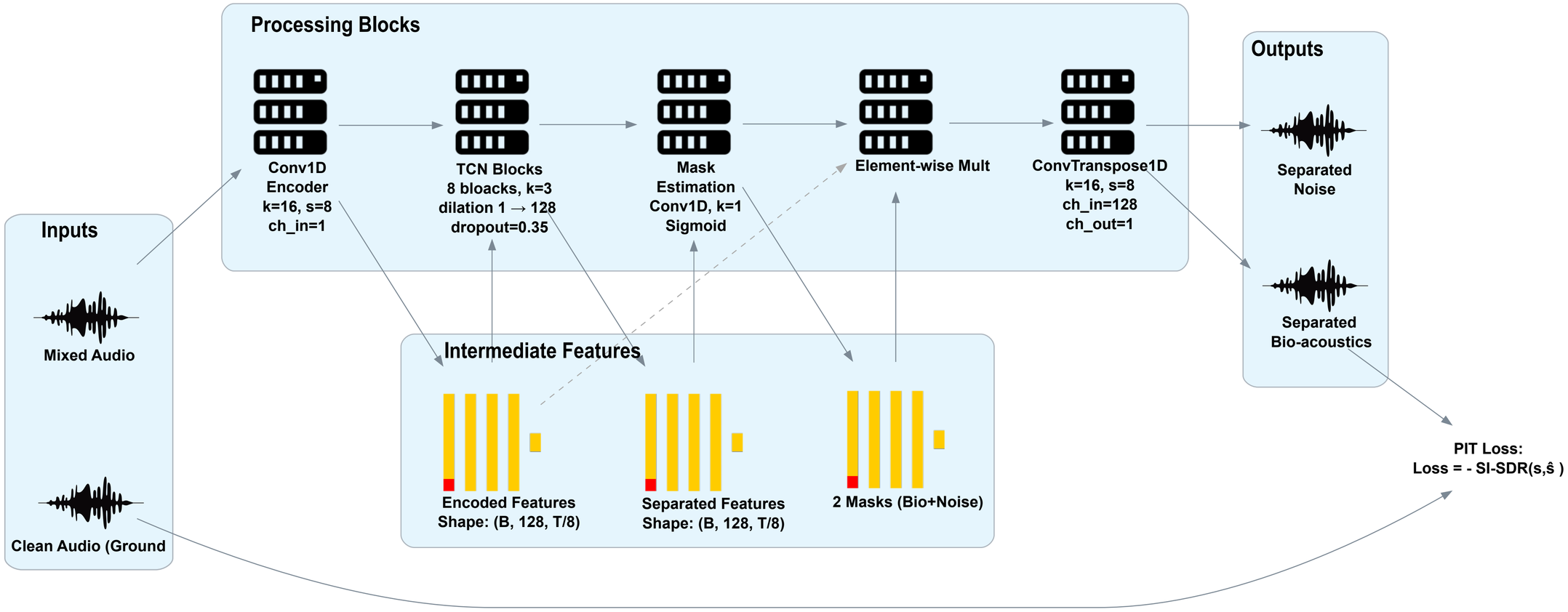
Processing Blocks
Conv1D
Encoder
k=16, s=8
ch_in=1
TCN Blocks
8 bloacks, k=3
dilation 1 → 128
dropout=0.35
Mask
Estimation
Conv1D, k=1
Sigmoid
Element-wise Mult
ConvTranspose1D
k=16, s=8
ch_in=128
ch_out=1
Outputs
Separated
Noise
Separated
Bio-acoustics
Inputs
Mixed Audio
Clean Audio (Ground
Intermediate Features
Encoded Features
Shape: (B, 128, T/8)
Separated Features
Shape: (B, 128, T/8)
2 Masks (Bio+Noise)
PIT Loss:
Loss = - SI-SDR(s,ŝ)